Assessment -9

Principal ComponentAnalysis

1. Load the dataset sent to your email into R, and verify that the packages ggplot2, ggfortify, FactoMineR, and factoextra are loaded.
2. Display the total number of rows and columns in the dataset.
3. Extract only the numeric columns, clean the data, and normalize it using the scale() function into an object named data_scaled.
4. Perform Principal Component Analysis on data_scaled using either the PCA() or prcomp() function, and print its summary.
5. Plot a scree plot showing the percentage of variance for each component with data labels, setting the y-axis limit from 0 to 60.
6. Create a PCA biplot with non-overlapping labels and a proper title.
7. Extract the PCA scores into a data frame named pca_scores and display the first six rows.
8. Draw a scatter plot of the first two principal components (PC1 vs. PC2) with blue circular points and your registration number in the title